

# Qingzhao Zhang

Assistant Professor, Electrical and Computer Engineering (ECE), University of Arizona

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## RESEARCH INTERESTS

System security (e.g., cyber-physical systems, autonomous driving), software security (e.g., program analysis, formal verification), AI security (e.g., adversarial attacks, robustness).

## EDUCATION

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| <b>University of Michigan</b><br><i>Ph.D. in Computer Science and Engineering</i> | <i>Sep 2019 – May 2025</i> |
| <b>University of Michigan</b><br><i>M.S. in Computer Science and Engineering</i>  | <i>Sep 2019 – Dec 2023</i> |
| <b>Shanghai Jiao Tong University</b><br><i>B.E. in Computer Engineering</i>       | <i>Sep 2015 – May 2019</i> |

## WORK EXPERIENCE

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| <b>Assistant Professor   University of Arizona, ECE</b><br>• Lead <i>Secure Autonomous Systems</i> (SAS) research group working on computer security and systems. We advance the security, resilience, and trustworthiness of emerging autonomous systems (e.g., self-driving vehicles, drones, and robots) through cross-layer design and analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <i>Aug 2025 - Present</i>   |
| <b>Research Assistant/Research Fellow   University of Michigan, EECS</b><br><i>Supervisor: Prof. Z. Morley Mao</i><br>• Research on software/system for cyber-physical system safety — <i>AVChecker</i> , the first traffic rule compliance checker on autonomous driving software; <i>SmtConf</i> , safety vetting of industrial control system configuration.<br>• Research on adversarial machine learning on cyber-physical systems — Adversarial attack and mitigation on trajectory prediction on autonomous driving; Analyzed data fabrication vulnerability on collaborative perception.<br>• Research on robustness of vehicular network — <i>RAO</i> , collaborative perception with asynchronous sensors.<br>• Research on robust perception algorithms of autonomous driving, and large language model security/efficiency. | <i>May 2020 - Aug 2025</i>  |
| <b>Software Engineer Intern   Google</b><br>• Designed and implemented formal verification solutions to enhance the security properties of an embedded system kernel (based on open-sourced Tock OS), involving modular model checking and theorem proving.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <i>May 2023 - July 2023</i> |
| <b>Software Engineer Intern   Google</b><br>• Designed and implemented static analysis checks based on Android Lint for Google's Android tests, which was deployed to assist Google developers to write high-quality unit tests.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <i>May 2022 - July 2022</i> |

## PUBLICATIONS

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| <b>Conference Papers</b><br>• [ACNS '26] Anrin Chakraborti, <b>Qingzhao Zhang</b> , Jingjia Peng, Z. Morley Mao, Michael K. Reiter. "Automatic Teller Machines for Offline Ecash", <i>The 24th International Conference on Applied Cryptography and Network Security</i> .<br>• [S&P '26] Jiarui Li, Joey Brewington, <b>Qingzhao Zhang</b> , Z. Morley Mao. "Banshee: Target Switch Attacks on Gimbal-Stabilized Visual Tracking Systems via Acoustic Injection", <i>The 47th IEEE Symposium on Security and Privacy</i> . |
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- [COLM '25] Shuowei Jin, Yongji Wu, Haizhong Zheng, **Qingzhao Zhang**, Matthew Lentz, Z Morley Mao, Atul Prakash, Feng Qian, Danyang Zhuo. “Plato: Plan to Efficiently Decode for Large Language Model Inference”, *Conference on Language Modeling*.
- [ICML '25] Shuowei Jin\*, Xueshen Liu\*, **Qingzhao Zhang**, Z. Morley Mao. “Compute Or Load KV Cache? Why Not Both?”, *Forty-Second International Conference on Machine Learning*.
- [ICLR '25] Minkyong Cho, Yulong Cao, Jiachen Sun, **Qingzhao Zhang**, Marco Pavone, Jeong Joon Park, Heng Yang, Z. Morley Mao. “Cocoon: Robust Multi-Modal Perception with Uncertainty-Aware Sensor Fusion”, *The International Conference on Learning Representations*.
- [USENIX Security '24] **Qingzhao Zhang**, Shuowei Jin, Ruiyang Zhu, Jiachen Sun, Xumiao Zhang, Qi Alfred Chen, Z. Morley Mao. “On Data Fabrication in Collaborative Vehicular Perception: Attacks and Countermeasures”, *Proceedings of the 33rd USENIX Security Symposium*.
- [ICLR '24] Jiachen Sun, Haizhong Zheng, **Qingzhao Zhang**, Atul Prakash, Z. Morley Mao, Chaowei Xiao. “CALICO: Self-Supervised Camera-LiDAR Contrastive Pre-training for BEV Perception”, *The International Conference on Learning Representations*.
- [MobiCom '23] **Qingzhao Zhang**\*, Xumiao Zhang\*, Ruiyang Zhu\*, Fan Bai, Mohammad Naserian, Z. Morley Mao. “Robust Real-time Multi-vehicle Collaboration on Asynchronous Sensors”, *Proceedings of the 29th Annual International Conference on Mobile Computing and Networking*.
- [RAID '22] **Qingzhao Zhang**, Xiao Zhu, Mu Zhang, Z. Morley Mao. “Automated Runtime Mitigation for Misconfiguration Vulnerabilities in Industrial Control Systems”, *Proceedings of the 25th International Symposium on Research in Attacks, Intrusions and Defenses*.
- [CVPR '22] **Qingzhao Zhang**, Shengtuo Hu, Jiachen Sun, Qi Alfred Chen, Z. Morley Mao. “On adversarial robustness of trajectory prediction for autonomous vehicles”, *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- [IV '22] Ze Zhang, Sanjay Sri Vallabh Singapuram, **Qingzhao Zhang**, David Ke Hong, Brandon Nguyen, Z. Morley Mao, Scott Mahlke, Qi Alfred Chen. “AVMaestro: A Centralized Policy Enforcement Framework for Safe Autonomous-driving Environments”, *IEEE Intelligent Vehicles Symposium*.
- [AsiaCCS '22] Shengtuo Hu, **Qingzhao Zhang**, André Weimerskirch, Z. Morley Mao. “GateKeeper: A Gateway-based Broadcast Authentication Protocol for the In-vehicle Ethernet”, *Proceedings of the ACM on Asia Conference on Computer and Communications Security*.
- [SIGMETRICS '21] **Qingzhao Zhang**, David Ke Hong, Ze Zhang, Qi Alfred Chen, Scott Mahlke, Z Morley Mao. “A systematic framework to identify violations of scenario-dependent driving rules in autonomous vehicle software”, *Proceedings of the ACM on Measurement and Analysis of Computing Systems*.
- [SANER '20] **Qingzhao Zhang**\*, Yizhuo Wang\*, Juanru Li, Siqi Ma. “Ethploit: From fuzzing to efficient exploit generation against smart contracts”, *IEEE 27th International Conference on Software Analysis, Evolution and Reengineering*.
- [FC '20] Tsz Hon Yuen, Shi-feng Sun, Joseph K Liu, Man Ho Au, Muhammed F Esgin, **Qingzhao Zhang**, Dawu Gu. “Ringct 3.0 for blockchain confidential transaction: Shorter size and stronger security”, *Financial Cryptography and Data Security: 24th International Conference*.

## Journal Papers

- [TDSC] Yanxue Jia, Shi-Feng Sun, Yuncong Zhang, **Qingzhao Zhang**, Ning Ding, Zhiqiang Liu, Joseph K Liu, Dawu Gu. “PBT: A New Privacy-Preserving Payment Protocol for Blockchain Transactions”, *IEEE Transactions on Dependable and Secure Computing*.

## Workshop and Short Papers

- [LAMPS '25] **Qingzhao Zhang**, Ziyang Xiong, Z. Morley Mao. “Safeguard is a Double-edged Sword: Denial-of-service Attack on Large Language Models”, *2nd ACM Workshop on Large AI Systems and Models with Privacy and Safety Analysis*.

- [VehicleSec '25] Jiarui Li, Joseph Brewington, **Qingzhao Zhang**, Z. Morley Mao. “WIP: Hijacking Attacks on UAV Follow-Me Systems in Realistic Scenarios”, *3rd USENIX Symposium on Vehicle Security and Privacy*.
- [CPSIoTSec '24] **Qingzhao Zhang**, Z. Morley Mao. “Stealthy Data Fabrication in Collaborative Vehicular Perception”, *the 6th Workshop on CPS and IoT Security*.
- [SRML '22] Jiachen Sun, **Qingzhao Zhang**, Bhavya Kailkhura, Zhiding Yu, Chaowei Xiao, Z. Morley Mao. “ModelNet40-C: a robustness benchmark for 3D point cloud recognition under corruption”, *Workshop on Socially Responsible Machine Learning*.
- [FEAST '20] Ze Zhang, **Qingzhao Zhang**, Brandon Nguyen, Sanjay Sri Vallabh Singapuram, Z Morley Mao, Scott Mahlke. “Automatic Feature Isolation in Network Protocol Software Implementations”, *Proceedings of the 2020 ACM Workshop on Forming an Ecosystem Around Software Transformation*.

## Preprints

- Yutong Liu, Chenyi Wang, Ming F. Li, **Qingzhao Zhang**. “Adversarial Trust Poisoning in Vehicular Collaborative Perception”, *arXiv.org e-Print archive*.
- **Qingzhao Zhang**, Runting Zhang, Z. Morley Mao. “From Stealthy Data Fabrication to Unsafe Driving: Realistic Scenario Attacks on Collaborative Perception”, *arXiv.org e-Print archive*.
- Shuo Ju, **Qingzhao Zhang**, Huashan Chen, Xuheng Wang, Haotang Li, Wanqian Zhang, Feng Liu, Kebin Peng, Sen He. “Still Camouflage, Moving Illusion: View-Induced Trajectory Manipulation in Autonomous Driving”, *arXiv.org e-Print archive*.
- **Qingzhao Zhang**, Shaocheng Luo, Z Morley Mao, Miroslav Pajic, Michael K Reiter. “SoK: How Sensor Attacks Disrupt Autonomous Vehicles: An End-to-end Analysis, Challenges, and Missed Threats”, *arXiv.org e-Print archive*.
- Fangzhou Wu, **Qingzhao Zhang**, Ati Priya Bajaj, Tiffany Bao, Ning Zhang, Ruoyu Wang, Chaowei Xiao. “Exploring the Limits of ChatGPT in Software Security Applications”, *arXiv.org e-Print archive*.
- R. Spencer Hallyburton, **Qingzhao Zhang**, Z. Morley Mao, Miroslav Pajic. “Partial-Information, Longitudinal Cyber Attacks on LiDAR in Autonomous Vehicles”, *arXiv.org e-Print archive*.
- Jiachen Sun, **Qingzhao Zhang**, Bhavya Kailkhura, Zhiding Yu, Chaowei Xiao, Z. Morley Mao. “Benchmarking Robustness of 3D Point Cloud Recognition Against Common Corruptions”, *arXiv.org e-Print archive*.

## TEACHING AND MENTORSHIP

### Course Instructor, University of Arizona

- Spring 2026, ECE 402C/502C Operating System Design.

### PhD Students, University of Arizona

- Yutong Liu, started in Fall 2025.

### Supervisor of Undergraduate Research, University of Michigan

2022

- Multidisciplinary Design Program (MDP) at University of Michigan. Lead cybersecurity research and supervised a team of four students.

### Research Mentorship, University of Michigan

2020-2024

- Research on program analysis, autonomous driving, adversarial machine learning.
- Mentees: Charles Ziegenbein Jr., Kevin Zhang, Andrew Wei, Xingyu Wang, Jingjia Peng, Ziyang Xiong, Runting Zhang.

## SERVICES

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- Conference Committee Member: S&P 2027, AsiaCCS 2027, CCS 2026/2025
  - Conference Reviewer: NeurIPS 2026, COLM 2026, CVPR 2026, ECCV 2026, ICLR 2026/2025, ICRA 2025, ACM MM 2024, IV 2024/2023
  - Journal Reviewer: TDSC, IoT-J, Sensors, TIFS, ITSM
  - Workshop Committee Member: RICSS 2024, RICSS 2025
  - Artifact Reviewer: Usenix Security 2022
  - Pre-review Task Force: NSDI 2025

## TALKS

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- Enhancing Security, Safety, and Reliability of Cyber-physical Systems  
*Presentation at Midwest Security Workshop* 11/16/2024
- Stealthy Data Fabrication in Collaborative Vehicular Perception  
*Presentation at 6th CPSIoTSec Workshop, co-located with CCS 2024* 10/18/2024  
*Poster at Midwest Security Workshop* 11/16/2024
- On data fabrication in collaborative vehicular perception: attacks and countermeasures  
*Poster at Athena institute (NSF AI Institute) annual showcase (best poster award)* 5/3/2023  
*Presentation at USENIX Security 2024* 8/16/2024
- Automated runtime mitigation for timing-based safety hazards in industrial controllers  
*Virtual presentation at RAID 2022* 10/27/2022
- On adversarial robustness of trajectory prediction for autonomous vehicles  
*Poster at Athena institute (NSF AI Institute) annual showcase* 8/18/2022  
*Poster at CVPR 2022* 6/23/2022
- Robustness of applications for autonomous vehicles  
*Presentation at Workshop on Future Automotive Research Datasets* 11/19/2021
- A systematic framework for checking driving rule compliance in autonomous vehicle software  
*Virtual presentation at SIGMETRICS 2021* 6/16/2021

## HONORS AND AWARDS

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- Student Travel Grant, Usenix Security 2024, SIGMETRICS 2021, CCS 2021 2019 – 2024
- Rackham Student Fellowship, University of Michigan 2019
- Academic Excellence Scholarship of SJTU (top 5%) 2016, 2017, 2018